MoodMimi: An Emotion-Aware Chatbot Prototype for Affective Computing

Jashlein Leanne T. Marquez

April 2026

Abstract

This technical report presents MoodMimi, a web-based emotion-aware chatbot prototype developed as a school requirement for the subject Current Trends and Topics in Computing. The project focuses on Affective Computing, a field of computing concerned with systems that can recognize, interpret, process, and respond to human emotions. MoodMimi was designed to demonstrate how emotion detection can be integrated into a chatbot interface to make digital interaction more human-centered, reflective, and emotionally aware. MoodMimi allows users to type short messages, analyzes the emotional tone of the text, and provides a supportive chatbot response based on the detected emotion. The prototype was developed using HTML, CSS, and JavaScript for the frontend, Python FastAPI for the backend, and SQLite for local mood history storage. For the main emotion detection model, the system uses boltuix/NeuroFeel from Hugging Face, a lightweight text classification model designed for short-text emotion detection. NeuroFeel is described as supporting 13 emotional categories, including happiness, sadness, anger, fear, love, surprise, neutral, confusion, desire, guilt, sarcasm, and shame. The system also includes a keyword-based fallback detector, a mascot-based user interface, an emotion confidence panel, mood history tracking, and a crisis safety layer for detecting potentially harmful or crisis-related messages. The prototype demonstrates how Affective Computing, Natural Language Processing, and Human-Computer Interaction can be combined to create a chatbot that does not only process user input, but also responds according to the emotional meaning behind the message.

Keywords: Affective Computing, Emotion Detection, Chatbot, Natural Language Processing, Human-Computer Interaction, FastAPI, Hugging Face, NeuroFeel

1 Introduction

Chatbots are increasingly used in websites, mobile applications, customer service platforms, education systems, and personal productivity tools. Traditional chatbots, however, usually focus on answering questions or following scripted conversation flows. While this allows them to respond to user input, they often fail to consider the emotional state behind the user's message. This limitation is important because communication is not only about words, but also about emotional tone, intention, and context.

MoodMimi was developed to explore this gap through the concept of Affective Computing. Affective Computing was introduced and popularized by Rosalind Picard and generally refers to computing that relates to, arises from, or influences emotions. It involves enabling systems to recognize and respond to human affective states such as happiness, sadness, anger, fear, confusion, and frustration. In the context of MoodMimi, Affective Computing is applied through text-based emotion detection. The system reads a user's message, classifies the likely emotional tone, and produces a response that reflects the detected emotion.

The project was inspired by simple conversational chatbot applications, but MoodMimi extends the idea by adding an emotional understanding layer. Instead of functioning only as a question-and-answer bot, MoodMimi acts as a small mascot companion that helps users identify what they might be feeling. The mascot interface was designed to make the application feel friendly, approachable, and less intimidating for users who may be expressing emotional thoughts.

This project does not aim to replace professional mental health support or provide clinical diagnosis. Rather, it serves as an educational prototype that demonstrates how current trends in computing, particularly Artificial Intelligence, Natural Language Processing,

Affective Computing, and Human-Computer Interaction, can be integrated into a functional software application.

2 Methodology

2.1 System Overview

MoodMimi follows a client-server architecture. The frontend is responsible for the visual interface, including the landing page, chat screen, mascot display, emotion panel, confidence score, reflection prompt, and mood history. The backend is responsible for receiving user messages, processing the emotional tone, generating a response, storing the result, and returning data to the frontend.

The system workflow begins when the user opens the application. The landing page introduces MoodMimi as an emotion-aware chatbot prototype. When the user clicks the “Talk to MoodMimi” button, the interface switches to the chatbot screen. The user can then type a message into the chat input field. JavaScript sends the message to the FastAPI backend through an API request. The backend checks the message, processes it through the emotion detection engine, generates a supportive response, saves the interaction in SQLite, and sends the result back to the frontend.

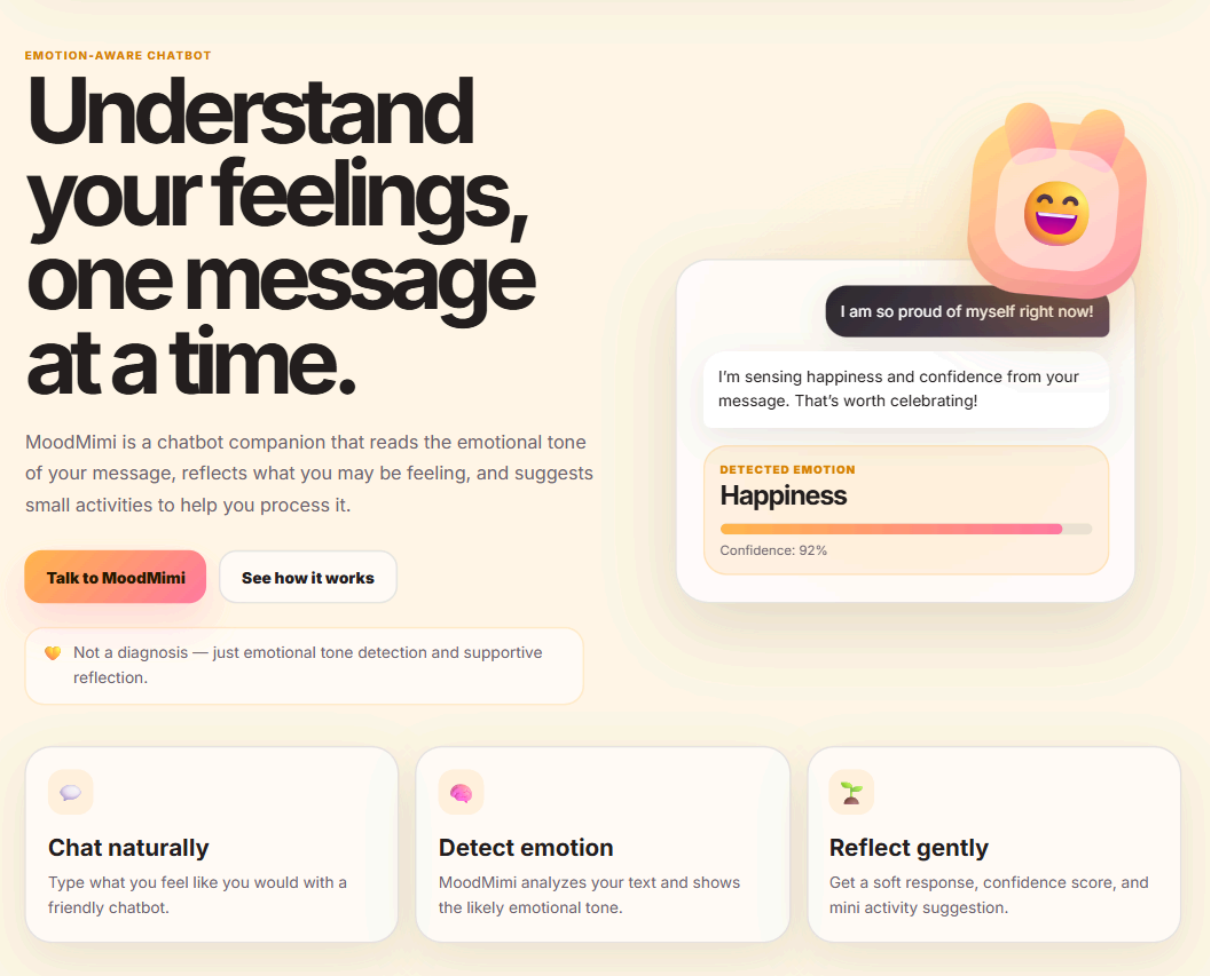

Figure 1.MoodMimi Landing Page

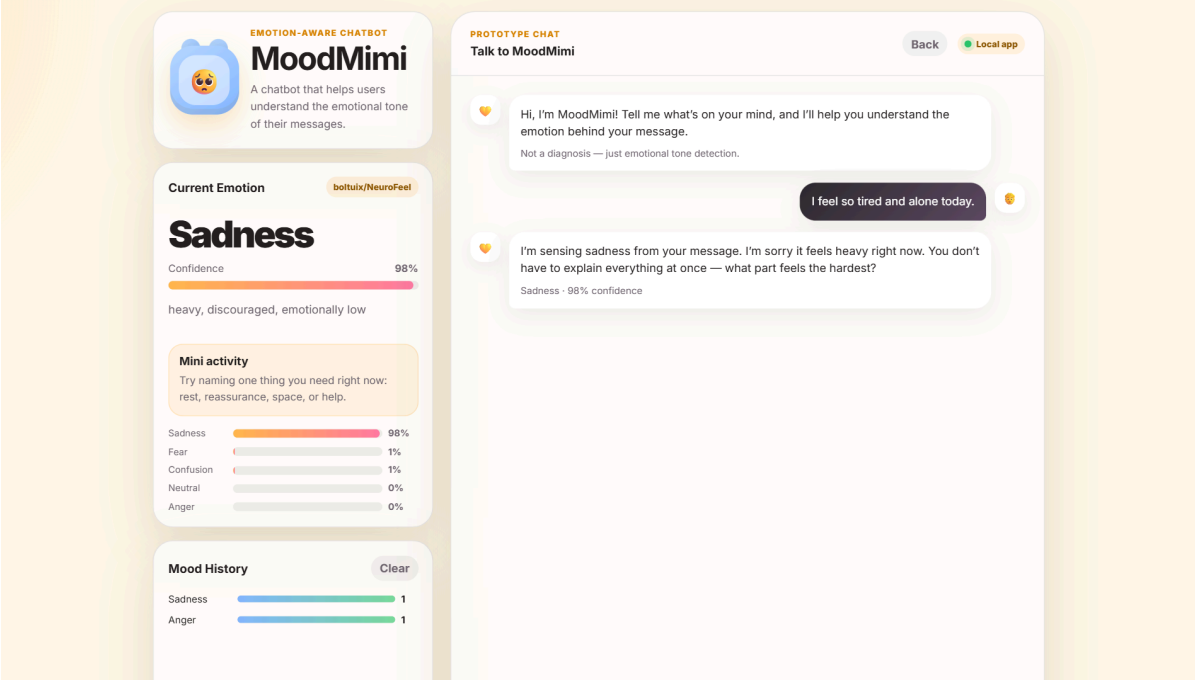

Figure 2.MoodMimi Chat

The returned result includes the chatbot reply, detected emotion, confidence score, emotional tone, suggested mini activity, model used, and timestamp. The frontend then updates the chat bubbles, current emotion card, confidence bar, mascot expression, and mood history.

2.2 Frontend Development

The frontend was developed using HTML, CSS, and JavaScript. HTML was used to define the structure of the landing page and chat interface. CSS was used to style the application with a soft, friendly visual design that matches the personality of MoodMimi. JavaScript was used to handle user interaction, button events, chat submission, API communication, screen switching, and dynamic UI updates.

The landing page introduces the purpose of the system and encourages the user to start a conversation. The chat page contains the main chatbot interface where the user sends messages and receives responses. The left panel displays the detected emotion,

confidence percentage, emotional tone, mini activity suggestion, and mood history. The mascot changes its visual state depending on the predicted emotion, helping communicate the emotional result in a more engaging and user-friendly way.

2.3 Backend Development

The backend was developed using Python FastAPI. FastAPI was used because it is suitable for building lightweight and fast API-based applications. It handles the main endpoints of the system, including the chat endpoint, mood history endpoint, and static file serving. The backend receives the user's message from the frontend, validates the input, passes the message to the emotion detection engine, and returns a structured response.

SQLite was used as the local database for storing chat records and emotion history. Each saved record includes the user message, chatbot reply, detected emotion, confidence score, tone description, suggestion, model used, and timestamp. This allows the system to display a simple mood history without requiring a separate external database server.

2.4 Emotion Detection Model

The main model used in MoodMimi is boltuix/NeuroFeel, a Hugging Face text classification model intended for short-text emotion detection. NeuroFeel is designed for short messages such as chatbot inputs, social media text, and feedback-style sentences. The model supports 13 emotion categories, which makes it more detailed than a simple positive, negative, or neutral sentiment analyzer.

The related boltuix emotions dataset contains 131,306 text entries labeled across 13 emotion categories, which supports the type of emotion classification needed for MoodMimi. This dataset and model direction are appropriate for the project because MoodMimi is designed to classify emotional tone from user-written text messages.

In the system, the backend loads the emotion detection pipeline and sends the user message to the model. The model returns emotion labels with confidence scores. The highest-scoring label is selected as the detected emotion. For example, a message such as “I am so proud of myself right now” may be classified as happiness, while a message such as “I feel so tired and alone today” may be classified as sadness or a related negative emotion.

To make the prototype more reliable during demonstrations, MoodMimi also includes a keyword-based fallback detector. If the transformer model fails to load due to internet connection, dependency issues, or Hugging Face download problems, the fallback detector still allows the system to classify basic emotions based on emotional keywords. Although this fallback approach is less accurate than the trained model, it ensures that the application remains functional.

2.5 Application of Affective Computing

Affective Computing is applied in MoodMimi through the system’s ability to interpret emotional meaning from user input and adjust the application’s response accordingly. The user’s message is not treated as plain text only. Instead, it becomes an affective signal that the system analyzes to infer the user’s possible emotional state.

The affective computing process in MoodMimi includes four main steps. First, the user expresses a message through natural language. Second, the system applies emotion detection to classify the message into an emotional category. Third, the chatbot generates a response that matches the detected emotion. Fourth, the interface displays visual and textual feedback, including the mascot reaction, confidence score, tone description, and reflection prompt.

This approach makes the chatbot more emotionally aware compared to a basic rule-based chatbot. For example, if the user expresses anger, MoodMimi responds with a calming tone. If the user expresses sadness, it responds with comfort and reflection. If the user expresses happiness, it responds with encouragement. Through this design, MoodMimi demonstrates how human-computer interaction can be improved by considering emotional context.

2.6 Safety Layer

Because the application deals with emotional input, MoodMimi includes a basic crisis safety layer. This layer checks for messages that may indicate self-harm or severe emotional distress. When such phrases are detected, the system does not continue with a normal chatbot response. Instead, it displays a supportive safety message and encourages the user to contact trusted people or professional support.

This feature is important because emotion-aware systems must be designed responsibly. MoodMimi is not a therapy tool, so it avoids giving clinical advice or diagnosis. The safety layer helps establish the boundaries of the system and reinforces that the prototype is intended only for emotional tone detection and educational demonstration.

3 Results and Discussion

MoodMimi successfully produced a working full-stack prototype that integrates frontend interaction, backend processing, emotion detection, response generation, and local history storage. The completed application includes a landing page, chatbot screen, mascot reaction system, emotion confidence card, mood history panel, and safety handling.

During prototype testing, the system was able to receive user messages, send them to the backend, process the emotional tone, and return structured results to the frontend. The frontend then updated the chat interface and emotion panel based on the result. Sample messages were used to check the behavior of the system. Positive messages were expected to produce emotions such as happiness or love, while negative messages were expected to produce sadness, anger, fear, shame, guilt, or confusion depending on the wording.

One strength of the system is that it provides more than just an emotion label. It also shows the confidence percentage, emotional tone interpretation, and a mini reflection activity. This makes the result more understandable for the user. For example, instead of only displaying “sadness,” MoodMimi may also describe the tone as discouraged or emotionally heavy and suggest a short reflective prompt.

The use of rule-based supportive responses also makes the chatbot safer and more predictable. Instead of relying on a fully generative AI model that may produce uncontrolled responses, MoodMimi uses predefined supportive replies based on the detected emotion. This is appropriate for a school prototype because the goal is to demonstrate emotion-aware interaction, not to create a clinical or therapeutic chatbot.

However, the project also has limitations. First, MoodMimi only detects emotion from text. It does not analyze voice tone, facial expression, or physiological signals. This means the system cannot fully capture emotional context, especially when the user’s message is sarcastic, vague, or lacking detail. Second, emotion classification is probabilistic and may not always reflect the user’s true feelings. Third, the fallback keyword detector is useful for functionality but has lower accuracy than a transformer-based model. Lastly, the system is not designed for real mental health treatment and should not be used as a replacement for professional help.

Despite these limitations, MoodMimi effectively demonstrates how Affective Computing can be applied in a practical software prototype. It combines current computing trends such as artificial intelligence, natural language processing, emotion recognition, and human-centered design into a simple but functional chatbot system.

4 Conclusion

MoodMimi is an emotion-aware chatbot prototype that demonstrates the application of Affective Computing in a web-based conversational system.

The system uses HTML, CSS, and JavaScript for the frontend, Python FastAPI for the backend, SQLite for local data storage, and the Hugging Face NeuroFeel model for emotion detection. MoodMimi classifies short text messages into emotional categories, displays confidence scores, updates a mascot interface, provides supportive chatbot responses, and stores recent emotion history.

The project shows that Affective Computing can make chatbot systems more emotionally aware and human-centered. By interpreting not only what the user says but also the possible feeling behind the message, MoodMimi creates a more reflective and engaging interaction. Although the prototype has limitations, such as text-only analysis and non-clinical use, it provides a meaningful demonstration of how emotion detection can be integrated into modern chatbot applications.

Future improvements may include voice-based tone detection, facial emotion recognition, multilingual emotion classification, better visual analytics for mood trends, online deployment, user accounts, improved safety handling, and more advanced response generation. These improvements could make MoodMimi more adaptive, personalized, and useful as an affective computing prototype.

References

boltuix. (n.d.). NeuroFeel. Hugging Face. https://huggingface.co/boltuix/NeuroFeel

Calvo, R. A., & D'Mello, S. (2010). Affect detection: An interdisciplinary review of models, methods, and their applications. IEEE Transactions on Affective Computing, 1(1), 18–37. https://doi.org/10.1109/T-AFFC.2010.1

Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A dataset of fine-grained emotions. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 4040–4054). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.372

Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent, Woebot: A randomized controlled trial. JMIR Mental Health, 4(2), Article e19. https://doi.org/10.2196/mental.7785

Holla, N., Prabal, P., H. U., N., Patil, N. R., & Nathaniel, D. E. (2025). A personalized mental health chatbot with emotion detection and content suggestion. Proceedings of SPIE, 13660, Article 136600J. https://doi.org/10.1117/12.3070958

Picard, R. W. (1997). Affective computing. MIT Press.